
Evaluating and inducing brain-like representations in self-supervised vision models

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Abstract

Recent work has shown that self-supervised vision models spontaneously develop brain-like representations when trained on natural images. We extend the analysis of Raugel et al. (1) along two complementary axes. First, we reproduce their key findings on temporal score and spatial score using DINO features on MEG and fMRI data, achieving a spatial score of $r = 0.910$ ($p < 0.001$). We additionally show that cross-subject averaging of MEG signals significantly boosts temporal encoding scores, improving the peak correlation from 0.09 to 0.185 on a model $85\times$ smaller than the original. Second, we introduce a *neural-guided training* approach in which MEG and fMRI brain signals act as biological regularizers during the training of a convolutional autoencoder, under two distinct architectural setups: an adversarial training scheme and a concatenation-based conditioning scheme. Our main finding is that biological guidance does not improve visual reconstruction quality, but the two setups exhibit markedly different convergence behaviors.

1 Introduction and Related Work

Recent work by Raugel et al. (1) demonstrates that self-supervised vision models, specifically DINOv3, exhibit a hierarchical organization that aligns with both the spatial structure of the visual cortex and its temporal dynamics. This alignment is typically quantified using an *encoding model* that maps network activations to brain signals (4; 5), where performance is measured via Pearson correlation. Beyond global scores, Raugel et al. define two key metrics: a *spatial score*, linking model depth to cortical hierarchy, and a *temporal score*, measuring the alignment between layers and the chronometry of MEG responses.

Despite these advances, it remains unclear if such alignment can be induced through direct supervision rather than merely observed as a spontaneous byproduct of training. In this work, we investigate two questions: **(i)** can these brain-model alignment results be reproduced in a lightweight setting with limited computational resources? and **(ii)** can we leverage brain signals as biological regularizers to steer model training? We first replicate the spatial and temporal analyses of Raugel et al. using DINO features, but by adding an averaging across subjects. We then introduce a neural-guided framework where MEG and fMRI signals act as supervision for a convolutional autoencoder through adversarial and concatenation-based schemes.

2 Datasets

Natural Scenes Dataset (NSD). We use the NSD (3), a 7T fMRI dataset of 8 subjects viewing 10,000 natural scenes. We compute encoding and spatial scores on a representative set of cortical regions of interest (ROIs) spanning V1 to IPS.

THINGS-MEG. We use THINGS-MEG (2), which contains MEG recordings from 4 subjects viewing 19,848 natural images from the THINGS object database. The MEG signal has shape (1, 271, 201) per trial in the training split and (12, 271, 201) in the test split, where 12 repetitions enable trial-averaging to improve SNR.

3 Temporal Score (MEG)

We reproduce the dynamic MEG encoding analysis of Raugel et al. (1) with a key methodological departure: while they average the encoding scores of four subjects, we first compute a *grand average* of the MEG signals across all subjects to maximize the Signal-to-Noise Ratio (SNR).

Figure 1 (left) shows that this signal-level aggregation suppresses idiosyncratic noise, allowing us to reach a peak correlation of 0.185—surpassing the 0.09 reported in the original study—even with a model $85\times$ smaller than DINOv3-Giant. Figure 1 (center) shows the peak correlation per DINO layer. We observe that intermediate layers (L5–L6) yield the highest alignment magnitude, suggesting they best capture the rich object representations characteristic of the ventral stream and high-level visual cortex (1). Notably, the correlation does not collapse in the deepest layers; instead, it maintains a stable, albeit lower, alignment. According to the framework of Raugel et al., these deeper representations are functionally distinct and specifically correspond to late-stage cognitive processes (300–400 ms), thereby reproducing the hierarchical temporal structure described in the original study (1).

SNR and Alignment. As shown in Figure 1 (right), the brain–model correlation (orange) closely tracks the MEG signal SNR (blue). This confirms that alignment is strictly noise-limited at the single-trial level, and justifies our use of signal averaging to recover stable representational similarity.

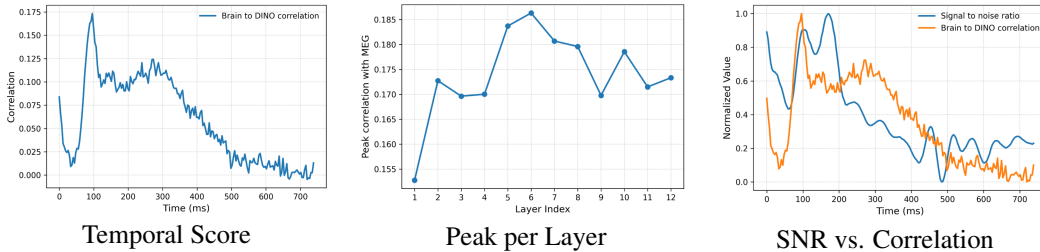


Figure 1: **Left:** Pearson correlation for grand average MEG. **Center:** Magnitude per layer (L5–L6 peak). **Right:** Correlation (orange) tracking SNR (blue), confirming noise limitations at single-subject levels.

4 Spatial Score (fMRI)

Encoding score. For each fMRI ROI, we train a ridge regression mapping DINO layer activations to BOLD responses. Unlike the MEG analysis, we focus here on a single representative subject (Subject 1) and average the BOLD responses across multiple occurrences of the same image to improve the signal-to-noise ratio. Pearson correlations are then computed on held-out images. Early visual areas (V1, V2) are best predicted by intermediate DINO layers (L5), while higher-order regions (OFA, FFA, OPA, PPA, EBA, IPS) are best predicted by the deepest layers (L11–L12), reproducing the hierarchical pattern of Raugel et al. (1). Finally, Figure 2 (far right) shows that DINO features systematically outperform a PCA-based raw pixel baseline across all ROIs. This gap widens in higher-order areas, confirming that DINO extracts biologically relevant representations beyond simple visual statistics.

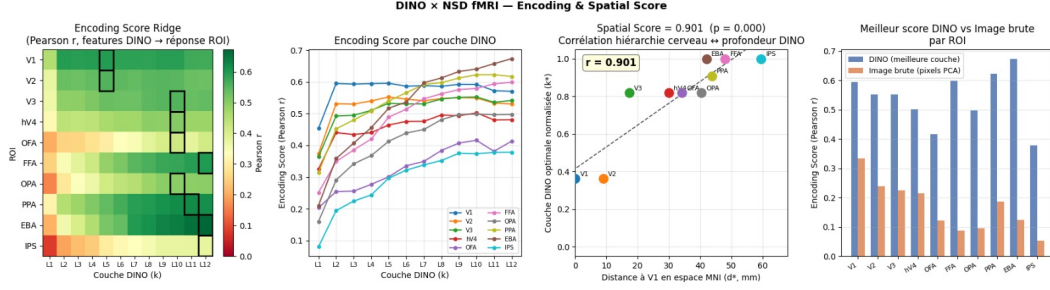


Figure 2: Encoding and spatial score analyses. **First:** Encoding score heatmap (Pearson r) across DINO layers and fMRI ROIs. **Second:** Encoding score per layer for each ROI. **Third:** Spatial score ($r = 0.910$, $p < 0.001$), showing the layer-to-distance hierarchy. **Fourth:** Comparison between the best DINO layer and raw image pixels (PCA baseline), highlighting the superior brain-predictive power of DINO features.

Spatial score. We compute the spatial score as the Pearson correlation between the optimal encoding layer k^* per ROI and its Euclidean distance from V1 in MNI space. Our reproduction yields $r = 0.910$ ($p < 0.001$), higher than the $r = 0.38$ of the original study. This discrepancy stems from a difference in spatial granularity: Raugel et al. operate at the voxel level (capturing biological noise), whereas our analysis averages within whole ROIs, acting as a spatial smoothing filter that reveals a cleaner layer-hierarchy mapping.

5 Neural-Guided Learning

5.1 Setup 1: Adversarial Training

A two-layer MLP **probe** h_ψ maps z to a predicted brain signal $\hat{y}$. The combined training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda \cdot \mathcal{L}_{\text{MSE}}, \quad (1)$$

where $\mathcal{L}_{\text{recon}} = \text{MSE}(\hat{x}, x)$ and $\mathcal{L}_{\text{MSE}} = \text{MSE}(\hat{y}, y)$ is the mean squared error on brain signals. Crucially, no `.detach()` is applied to z : the brain gradient back-propagates through the probe directly into the encoder, steering the latent space toward brain-predictive features.

5.2 Setup 2: Concatenation Training

Instead of a probe, the brain signal c (MEG or fMRI embedding) is directly **concatenated** with the latent code to form $[z; c]$, which is then passed to the decoder. The decoder is the only component trained with access to brain information; the encoder receives no brain gradient. The objective reduces to:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} = \text{MSE}(\hat{x}, x). \quad (2)$$

This setup tests whether conditioning the decoder on brain signals – rather than regularizing the encoder – can improve reconstruction quality.

5.3 Results

Figure 3 reports the validation reconstruction loss (MSE) for both setups. We observe that biological guidance consistently fails to improve visual reconstruction compared to the baseline (blue). In the **Adversarial setup**, the inclusion of $\mathcal{L}_{\text{brain}}$ raises the MSE floor, revealing a *representational competition*: the encoder is forced to prioritize biologically-relevant features (e.g., semantic invariants) at the expense of the low-level pixel information required for high-fidelity decoding. In the **Concatenation setup**, providing brain embeddings to the decoder does not assist reconstruction, likely because the high variance of single-trial neural data acts as a noisy prior rather than a helpful constraint.

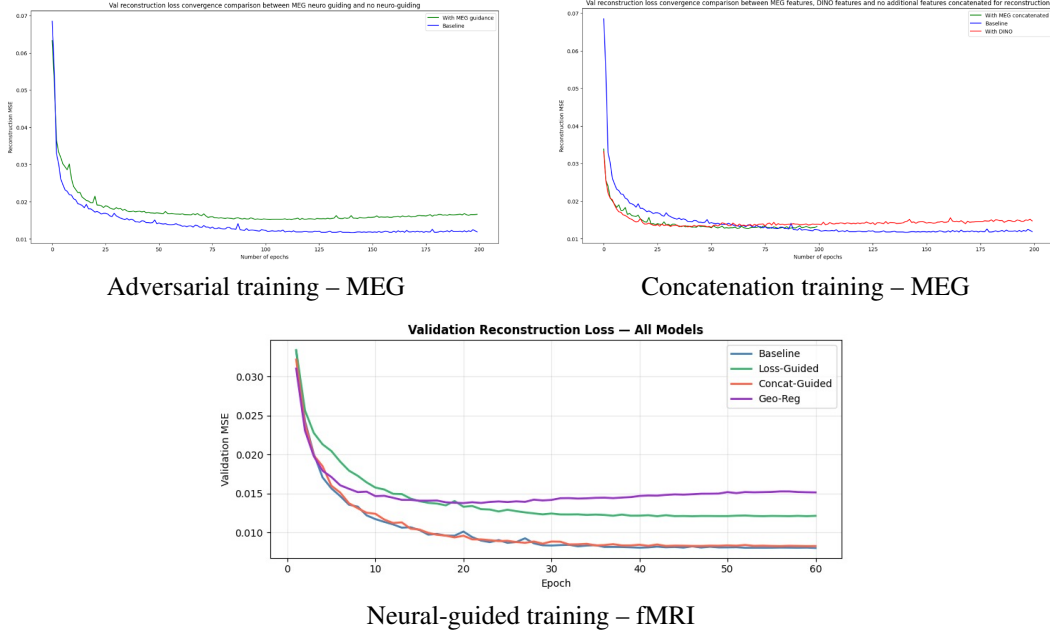


Figure 3: Validation reconstruction loss (MSE) for the two neural-guided setups on MEG (top) and fMRI (bottom). In each plot, the baseline (reconstruction-only) is shown in blue; the brain-guided condition is shown in green/red.

6 Discussion

Spontaneous vs. Induced Alignment. Our findings highlight a key distinction from Raugel et al. (1): while brain-like hierarchies emerge spontaneously in large-scale models trained on natural images, *actively forcing* this alignment in smaller models creates a performance bottleneck. This suggests that "seeing like a human" and "reconstructing pixels" are partially competing objectives at limited scales.

Limitations. The failure of neural-guided training to outperform the baseline stems from two primary constraints. First, the **hyperparameter space** for biological regularization (e.g., λ , learning rates) is vast and computationally expensive to tune, making optimal alignment difficult to reach. Second, while neural guidance might succeed in an idealized setting, our **model capacity** is likely insufficient: the network may lack the representational flexibility to satisfy both visual reconstruction and complex biological constraints simultaneously. Additionally, the high noise floor of single-trial MEG data further hinders the stability of the biological gradient.

7 Conclusion

This project explored brain-model convergence from two complementary angles. Our reproduction of Raugel et al.’s key findings confirms that DINO’s representational hierarchy faithfully mirrors cortical organization ($r = 0.910$), and demonstrates that cross-subject averaging of MEG signals substantially improves temporal encoding scores on a lightweight model. The neural-guided learning experiments show that, regardless of setup, biological guidance does not improve visual reconstruction quality – though the adversarial and concatenation schemes exhibit distinct failure modes worth exploring in future work.

References

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